EXPERIMENT 2

PROGRAM 1

2.1.1//Roots of quadratic equarion

ALGORITHM:

Step 1: Start
Step 2: Read coefficients a, b, and c
Step 3: Calculate discriminant

$$\boldsymbol{D = b^2 - 4ac}$$

Step 4: Check the value of D

- **If D > 0 (Real and different roots)**
  - **Compute**

$$\boldsymbol{root1 = \frac{-b + \sqrt{D}}{2a}}$$
$$\boldsymbol{root2 = \frac{-b - \sqrt{D}}{2a}}$$

  - **Print root1 and root2 (2 decimal places)**
- **Else if D == 0 (Real and same roots)**
  - **Compute**

$$\boldsymbol{root = \frac{-b}{2a}}$$

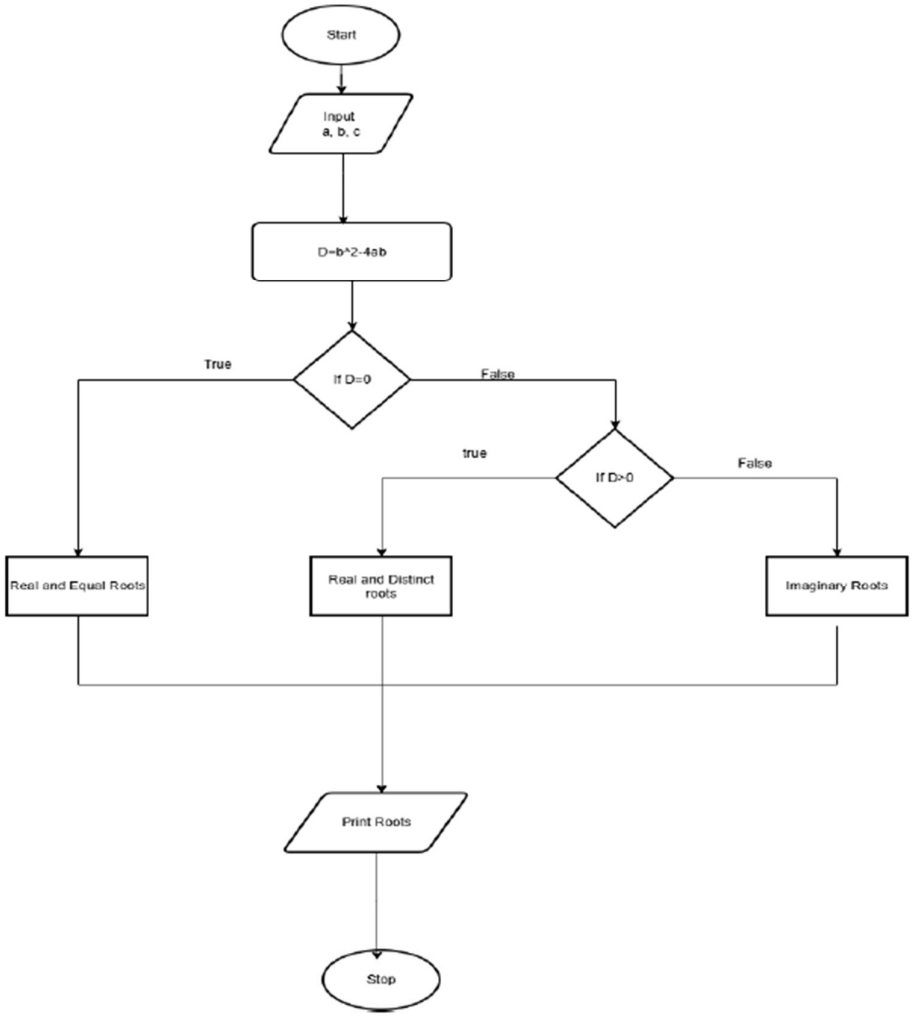
  - **Print root1 = root2 = root**
- **Else (D < 0) (Imaginary roots)**
  - **Compute**

$$\boldsymbol{realPart = \frac{-b}{2a}}$$
$$\boldsymbol{imaginaryPart = \frac{\sqrt{-D}}{2a}}$$

  - **Print**
    - **root1 = realPart + imaginaryPart i**
    - **root2 = realPart - imaginaryPart i**

Step 5: Stop

FLOWCHART:



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2.1.1. Roots of a Quadratic Equation

03:11

- If $D > 0$: Roots are real and different
- If $D = 0$: Roots are real and the same
- If $D < 0$: Roots are imaginary

Input Format:

- Three space-separated integers representing the coefficients a, b, and c, respectively.

Output Format:

- If roots are real and different, print:

```
root1 = <Root1>
root2 = <Root2>
```

- If roots are the same, print:

```
root1 = root2 = <Root1>
```

- If roots are imaginary, print:

```
root1 = <RealPart>+<ImaginaryPart>i
root2 = <RealPart>-<ImaginaryPart>i
```

- All values should be formatted to two decimal places.

Sample Test Cases +

Explorer

quadratic... Submit

```python
# Write your code here...
a,b,c = map(float,input().split())#space separated

D = (b * b) - 4 * a * c
sqrD = D ** 0.5

if D > 0:
    root1 = (-b + sqrD)/(2*a)
    root2 = (-b - sqrD)/(2*a)
    print(f"root1 = {root1 :.2f}")
    print(f"root2 = {root2 :.2f}")
```

Debugger

Average time 0.005 s 5.00 ms

Maximum time 0.009 s 9.00 ms

3 out of 3 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 9 ms Debug

Expected output	Actual output
1 -5 6	1 -5 6
root1 = 3.00	root1 = 3.00
root2 = 2.00	root2 = 2.00

Test case 2 4 ms

Terminal Test cases

Prev Reset Submit Next

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Input Format:

- Three space-separated integers representing the coefficients a, b, and c, respectively.

Output Format:

- If roots are real and different, print:

```
root1 = <Root1>
root2 = <Root2>
```

- If roots are the same, print:

```
root1 = root2 = <Root1>
```

- If roots are imaginary, print:

```
root1 = <RealPart>+<ImaginaryPart>i
root2 = <RealPart>-<ImaginaryPart>i
```

- All values should be formatted to two decimal places.

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quadratic... Submit

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D = (b * b) - 4 * a * c
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if D > 0:
    root1 = (-b + sqrD)/(2*a)
    root2 = (-b - sqrD)/(2*a)
    print(f"root1 = {root1 :.2f}")
    print(f"root2 = {root2 :.2f}")

elif D == 0:
    root = -b / (2*a)
    print(f"root1 = root2 = {root:.2f}")

else:
    real_part = (-b) / (2*a)
    imaginary_part = sqrD / (2*a)

    print(f"root1 = {real_part :.2f}+{imaginary_part.imag :.2f}i")
    print(f"root2 = {real_part :.2f}-{imaginary_part.imag :.2f}i")
```

Debugger

Terminal Test cases

Prev Reset Submit Next